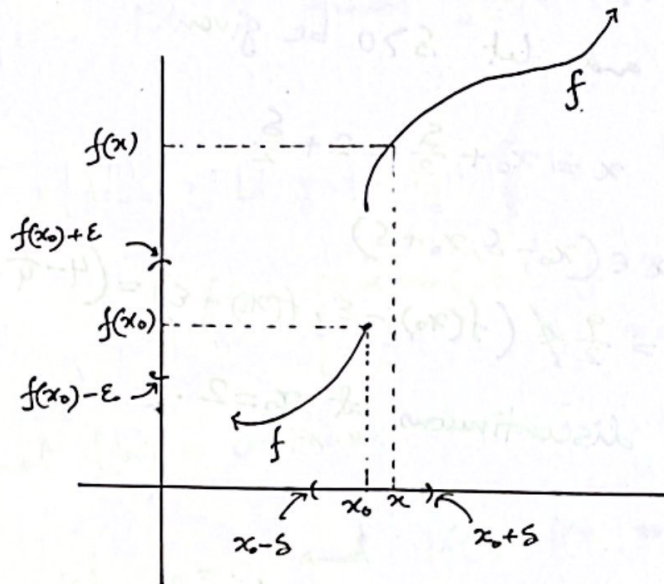


ϵ - δ definition

Continuity

We first define discontinuity of a function at a point.

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ is defined as the subset $\{(x, f(x)): x \in \mathbb{R}\}$ of the plane $\mathbb{R}^2$. Let $x_0 \in \mathbb{R}$ and if f is discontinuous at x_0 , then the graph of f is broken at the point $(x_0, f(x_0))$.



Observe in the figure that we can choose some ϵ -neighbourhood $(f(x_0) - \epsilon, f(x_0) + \epsilon)$ of $f(x_0)$, such that if we consider any δ -neighbourhood $(x_0 - \delta, x_0 + \delta)$ of x_0 , then we can find at least one element x in $(x_0 - \delta, x_0 + \delta)$ such that $f(x) \notin (f(x_0) - \epsilon, f(x_0) + \epsilon)$.

Definition of discontinuity

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is said to be discontinuous at a point $x_0 \in \mathbb{R}$ if there exists $\epsilon > 0$ such that for every $\delta > 0$ there exists $x \in (x_0 - \delta, x_0 + \delta)$ but $f(x) \notin (f(x_0) - \epsilon, f(x_0) + \epsilon)$.

Example: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined

$$f(x) = \begin{cases} x^2 & \text{if } x \leq 2 \\ \frac{9}{2} & \text{if } x > 2 \end{cases}$$

Does the above function continuous at $x_0 = 2$?

Solⁿ: Let $\varepsilon = \frac{1}{4}$ and let $\delta > 0$ be given.

Then define $x = x_0 + \frac{\delta}{2} = 2 + \frac{\delta}{2}$

Note that $x \in (x_0 - \delta, x_0 + \delta)$

But $f(x) = \frac{9}{2} \notin (f(x_0) - \varepsilon, f(x_0) + \varepsilon) = (4 - \frac{1}{4}, 4 + \frac{1}{4})$

Hence f is discontinuous at $x_0 = 2$.

Problem: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be defined

$$f(x) = \begin{cases} 0 & \text{if } x \in \mathbb{Q} \\ 1 & \text{if } x \notin \mathbb{Q} \end{cases}$$

Does $f(x)$ discontinuous at every element of $\mathbb{R}$? prove or disprove it.

Definition of Continuity (ϵ - δ definition)

A function $f: \mathbb{R} \rightarrow \mathbb{R}$ is continuous at a point $x_0 \in \mathbb{R}$ if $\forall \epsilon > 0 \exists \delta > 0$ such that whenever $|x - x_0| < \delta \Rightarrow |f(x) - f(x_0)| < \epsilon$.

Note that $|x - x_0| < \delta$ means $x \in (x_0 - \delta, x_0 + \delta)$ and $|f(x) - f(x_0)| < \epsilon$ means $f(x) \in (f(x_0) - \epsilon, f(x_0) + \epsilon)$.

Example: Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be given by

$$f(x) = \begin{cases} 2x \sin(1/x) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$$

Show that $f(x)$ is continuous at $x = 0$.

Sol. Here let $x_0 = 0$ and $|f(x) - f(x_0)| = |2x \sin(1/x) - 0| \leq |2x| = 2|x - x_0|$

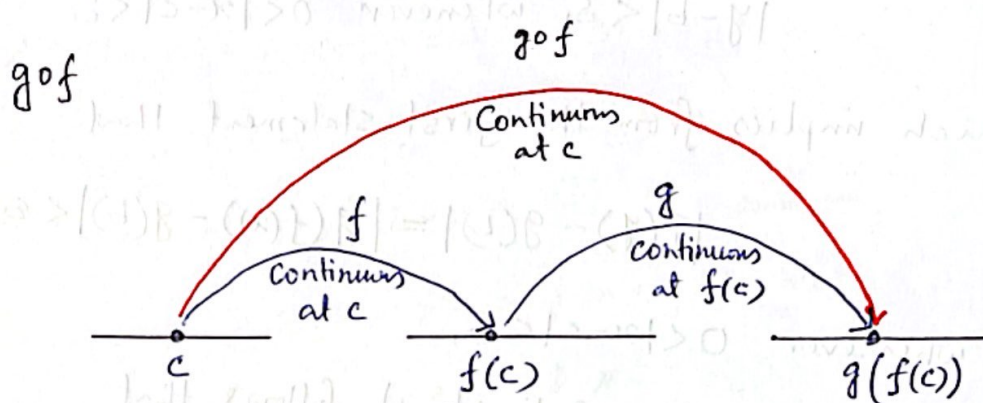
Since $|x - x_0| < \delta$

let ϵ be given. Choose any $\delta > 0$ s.t. $\delta \leq \frac{\epsilon}{2}$

Then $|f(x) - f(x_0)| < 2\delta \leq \epsilon$ whenever $|x - x_0| < \delta$.

This shows that f is continuous at $x_0 = 0$. *

Compositions of Functions



Theorem 1:

If f is continuous at c and g is continuous at $f(c)$, then the composition $g \circ f$ is continuous at c .

Theorem 2:

If $\lim_{x \rightarrow c} f(x) = b$ and g is continuous at the point b , then

$$\lim_{x \rightarrow c} g(f(x)) = g(b).$$

Proof: Let $\varepsilon > 0$ be given. Since g is continuous at b , there exist a number $\delta_1 > 0$ such that

$$|g(y) - g(b)| < \varepsilon \text{ whenever } 0 < |y - b| < \delta_1$$

Here, [say $\lim_{y \rightarrow b} g(y) = g(b)$ since g is continuous at $y = b$].

Since $\lim_{x \rightarrow c} f(x) = b$, there exists a $\delta > 0$ s.t.

$$|f(x) - b| < \delta_1 \text{ whenever } 0 < |x - c| < \delta$$

~~g ∘ f is continuous at c~~

Cont. ...

If we say $y = f(x)$, then we have

$$|y - b| < \delta, \text{ whenever } 0 < |x - c| < \delta$$

which implies from the first statement that

$$|g(y) - g(b)| = |g(f(x)) - g(b)| < \varepsilon$$

whenever $0 < |x - c| < \delta$.

From the definition of limit, it follows that

$$\lim_{x \rightarrow c} g(f(x)) = g(b).$$

Example 1 Estimate : $\lim_{x \rightarrow \pi/2} \cos(2x + \sin(\frac{3\pi}{2} + x))$

Sol. $\lim_{x \rightarrow \pi/2} \cos(2x + \sin(\frac{3\pi}{2} + x))$

$$= \lim_{x \rightarrow \pi/2} \cos\left(\lim_{x \rightarrow \pi/2} 2x + \lim_{x \rightarrow \pi/2} \sin(\frac{3\pi}{2} + x)\right)$$

$$= \cos(\pi + \sin 2\pi)$$

$$= \cos \pi$$

$$= -1.$$

✗

$$\frac{3\pi}{2} + \frac{\pi}{2} = \frac{4\pi}{2}$$

Order of a set

Let S be a non-empty set. An order on a set S is a relation denoted by " $<$ " with the following properties:

1. If $x \in S$ and $y \in S$ then one and only one of the statements $x < y$, $x > y$ and $x = y$ is true.
2. If $x, y, z \in S$ and if $x < y$, $y < z$ then $x < z$.

* A set S is said to be ordered if an order is defined on S .

Bounded sets

Let S be an ordered set and $E \subseteq S$. If there exists a $b \in S$ such that $x \leq b \forall x \in E$ then E is bounded above and b is known as an upper bound of E .

~~if there~~ If $\exists p \in S$ such that $x \geq p \forall x \in E$ then p is the lower bound of E .

Least upper bound (Supremum)

Suppose S is an ordered set, $E \subseteq S$ and E is bounded above. Suppose $\exists a \in S$ such that

- i. a is an upper bound of E
- ii. If $g < a$, then g is not an upper bound of E .

Then a is called a least upper bound or supremum of E .

i.e. $\sup E = a$.

Greatest lower bound (infimum)

Suppose S is an ordered set, $E \subseteq S$ and E is bounded below.

Suppose $\exists k \in S$ such that

i. k is a lower bound of E

ii. If $q > k$, then q is not a lower bound of E .

Then k is called a greatest lower bound or infimum of E .

i.e. $\inf E = k$.

Note: The supremum and the infimum are not necessarily members of E .

e.g. let $E = \{1, \frac{1}{2}, \frac{1}{3}, \frac{1}{4}, \frac{1}{5}, \dots\} = \{\frac{1}{n}\}$

then the $\sup E = 1 \in E$ and the $\inf E = 0 \notin E$.

Neighborhoods:

A neighborhood of a point $p \in \mathbb{R}$ is an interval of the form $(p-\delta, p+\delta)$ where $\delta > 0$ for any positive real number.

Thus, the neighborhood consists of all points distance less than δ from p .

$$(p-\delta, p+\delta) = \{x \in \mathbb{R} : |x-p| < \delta\}$$

e.g. say $\delta = 0.5$

so, $(1.5, 2.5)$ is a neighborhood of 2.

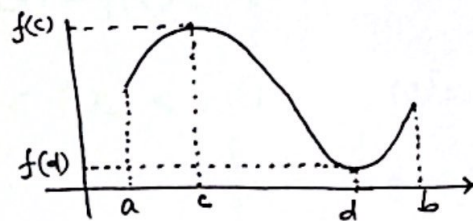
Continuous Function (ϵ - δ definition)

A function f is continuous at a point c if $\forall \epsilon > 0 \exists \delta > 0$ such that whenever $|x - c| < \delta \Rightarrow |f(x) - f(c)| < \epsilon$.

Extreme Value Theorem:

If a real-valued function f is continuous on the closed and bounded interval $[a, b]$, then f must attain a maximum and a minimum, each at least once. That is, there exist numbers c and d in $[a, b]$ such that

$$f(c) \leq f(x) \leq f(d) \quad \forall x \in [a, b].$$



Boundedness Theorem:

If f is continuous on the closed, bounded interval $[a, b]$, then f is bounded above and below in that interval.

That is, there exist numbers m and M such that

$$m \leq f(x) \leq M \quad \forall x \in [a, b].$$

* This does not say that M and m are necessarily the maximum and minimum values of f on the interval $[a, b]$.

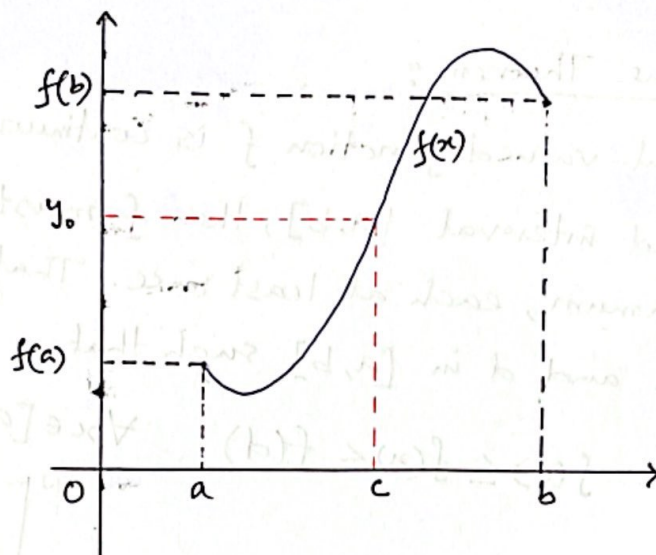
e.g. 1. $f(x) = x \quad x \in [0, \infty)$ is not bounded from above.

2. $f(x) = \frac{x}{1+x} \quad x \in [0, \infty)$ is bounded from below but does not attain its least upper bound 1.

The Intermediate Value Theorem

Statement:

If f is a continuous function on a closed interval $[a, b]$, and if y_0 is any value between $f(a)$ and $f(b)$, then $y_0 = f(c)$ for some c in $[a, b]$.



Importance: The intermediate value theorem tells us that if f is continuous, then any interval on which f changes sign contains a zero of the function. Somewhere between a point where a continuous function is positive and a second point where it is negative, the function must be equal to zero.

Problem: Show that there is a root of the eqⁿ, $x^3 - x - 1 = 0$ between 1 and 2.

Sol. Let, $f(x) = x^3 - x - 1$, $f(1) = 1 - 1 - 1 = -1 < 0$ and $f(2) = 5 > 0$

Since f is a polynomial, so it is continuous.

By IVT, we say that there is a zero of f betⁿ 1 and 2.

say, $y_0 = 0$ is a value betⁿ $f(1)$ and $f(2)$.

*

Proof of IVT:

Let f be a continuous function on the closed interval $[a, b]$.

Assume ~~that~~ without loss of generality $f(a) < K < f(b)$.

We define a set S s.t. $S = \{x \in [a, b] \mid f(x) < K\}$

Since $f(a) \leq K$, S is nonempty and bounded above by b .

Let $c = \sup S$

Now we show that $f(c) = K$

For $f(c) \geq K$

~~Suppose~~ Suppose $f(c) < K$. Since f is continuous at c , $\exists \delta > 0$ s.t.

$$|x - c| < \delta \Rightarrow |f(x) - f(c)| < K - f(c) \quad \text{say } \epsilon = K - f(c)$$

$$\Rightarrow -(K - f(c)) < f(x) - f(c) < K - f(c)$$

$$\Rightarrow f(c) - K + f(c) < f(x) < f(c) + K - f(c) = K$$

So, $f(x) < K$

So, for some $x > c$, $f(x) < K$, hence $x \in S$, which contradicts the fact that c is the least upper bound of S .

So, $f(c) \geq K$.

For $f(c) \leq K$

Suppose $f(c) > K$. Since f is continuous at c , $\exists \delta > 0$ s.t.

$$|x - c| < \delta \Rightarrow f(x) > K \quad \text{[same way from above inequality]}$$

So, for all $x \in (c - \delta, c)$, $f(x) > K$ which means no points in that interval belong to S . But then c is not the least upper bound which is contradiction.

Therefore, $f(c) = K$.

□